An 88.8dB-SNDR 6-MS/s Pipelined SAR ADC with A Closed-Loop Dynamic Amplifier Featuring Highly-Linear Full-Scale Output Swing

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Abstract

This paper presents a pipelined SAR ADC with a two-stage closed-loop residue amplifier (RA). The proposed RA incorporates a floating gm-ratio-based amplifier (FGA) as the first stage and a charge-pump-assisted cascoded floating inverter amplifier (CP-FIA) as the output stage. It enables fast-slewing, energy-efficient, robust, and highly-linear amplification, with a THD of <-100 dB at full-scale output swing. A system-level chopping-assisted auto-zero (AZ) is adopted to eliminate the RA's flicker noise and offset without additional thermal noise penalty. The prototype achieves 88.8dB SNDR at 6MS/s, demonstrating the best FoMs of 183dB among ADCs with >5MS/s conversion rate.

Introduction

The pipelined SAR ADC is well-suited for emerging applications requiring high resolution and low measurement time. For instance, in medical imaging, its low noise is crucial for image quality. Even with low-frequency input signals (less than tens of kHz for X-ray/CT), the fast sampling rate is highly desired for efficient multiplexing, thus reducing patients' exposure time. The key to high performance lies in the RA, which attenuates the backend noise and ensures conversion accuracy. Compared to the open-loop amplifier that often requires complex calibration [1]–[3], the closed-loop one is favored thanks to its accurate inter-stage gain and good linearity. The conventional closed-loop OTA exhibits robust performance against PVT variations [4]–[7], but it consumes static power. The ring amplifier offers high speed and slew efficiency [8]–[12]. However, it typically requires careful dead-zone tuning for stability. Owing to the self-quenching mechanism, the FIA-based amplifier realizes both high energy efficiency and good stability [13], [14]. Nevertheless, the reduced supply voltage during amplification limits its speed and linear output swing. To provide a fast and linear residue amplification while maintaining robustness and energy efficiency, this work proposes a novel closed-loop RA design.

Architecture and Circuit Implementation

The proposed two-stage closed-loop RA with 1st-stage FGA and 2nd-stage CP-FIA is shown in Fig.2. With the floating 1/gm load, the 1st-stage FGA's output impedance is largely reduced, offering a high non-dominant pole frequency, and thus, it enhances the amplification speed and stability. The gm ratio is designed as 8:1, leading to a PVT-robust FGA gain of 8. It also reduces the loop gain variation of the proposed RA. The 2nd-stage CP-FIA consists of a charge pump-assisted reservoir capacitor and a cascoded inverter amplifier with split input biasing. As shown in Fig.3, during the reset phase, the two reservoir capacitors C_{R2} are both charged to 1.8V, and the outputs are reset to 0.9V CM voltage. During the amplification phase, the two capacitors are connected in series, generating a doubled supply voltage domain for the cascoded inverter amplifier, thus offering higher slew current and larger output swing. The floating voltage domain offers two key merits: 1) no CM current flows into output nodes, ensuring a stable output CM voltage of 0.9V [15]; 2) the boosted floating voltage domain automatically centers around the pre-defined CM, i.e., the supply rails are shifted to 2.7V and -0.9V at the beginning of the amplification; it balances the strengths of PMOS/NMOS and ensures amplification linearity. The input and cascode pairs are split-biased to the strong inversion region using the embedded capacitor C_{BIAS} when the amplification starts. With the discharging of stacked reservoir capacitors, the input pair shifts to the subthreshold region, ensuring good energy efficiency and low noise bandwidth.

To guarantee the reliability of the proposed CP-FIA, sophisticated designs are required to handle the beyond-the-rail voltages. During the amplification phase, the drain voltages of reset switches M_{RP} and M_{RN} are respectively greater than 1.8V and less than 0V, making it difficult to be fully turned off. To mitigate potential channel and junction current leakages, this design shorts both the gate and bulk of reset switches to the drain node during amplification. As a result, the proposed CP-FIA can operate under beyond-the-rail supply voltages with only standard CMOS processes. Overall, the proposed RA realizes a THD of <-100dB with a full-scale output swing (Fig.4). A stable output CM voltage and >60 degrees phase margin can be guaranteed over temperature and process corner variations, demonstrating the robustness of the proposed RA.

Fig.5 depicts the simplified schematic of the proposed pipelined SAR ADC. It comprises a 6b 1st-stage SAR quantizer, a 12b backend SAR quantizer, and the proposed RA. The RA's boosted output swing enables a large inter-stage gain of 32, relaxing the backend quantizer's noise requirement and leading to a low system-level power consumption.

To mitigate the RA's flicker noise and offset impact, the embedded capacitors C_{BIAS} for split biasing are reused to perform output AZ for the 1st-stage FGA. However, in the conventional AZ, additional thermal noise will be introduced by the C_{BIAS} sampling [7], [9] (Fig.6). This work adopts the system-level chopping to provide twofold benefits. First, it further reduces the residue flicker noise after AZ. Besides, it mitigates the AZ-induced thermal noise. AZ is performed in the (2n)-th cycle, storing the thermal noise on C_{BIAS}. This noise, together with the residue voltage, will be quantized by the backend SAR ADC in two consecutive cycles. Note that the AZ is only performed in the even cycle, and thus, the two consecutive conversion results contain the same sampled noise with the inverted input signal. As a result, after the output averaging, the AZ-induced thermal noise can be canceled. This technique is highly effective for targeted applications that require high-precision and low-latency readout for low-to-medium-frequency inputs. Measurement shows that the ADC thermal noise is reduced by 4.3dB due to system-level chopping (1.3dB from AZ-induced thermal noise cancellation and 3dB from noise averaging).

Measurement Results

The prototype ADC in 180nm CMOS occupies an active area of 0.237mm^2 (Fig.10). One-time off-chip foreground calibration is performed to address the capacitor mismatches before all measurements. With a low input frequency of 10kHz, the measured SFDR and SNDR are 107.1dB and 89.2dB, respectively. With 1MHz input, the measured SFDR and SNDR are 100.5dB and 88.8dB, respectively. The measured DR is 89.6dB. The ADC core consumes 1.15mW at 6MS/s. To verify the robustness of the proposed ADC, 4 chips are tested over -20~80℃ temperature range, demonstrating a <2dB SNDR variation. Table I summarizes the performance of this pipelined SAR ADC and compares it with state-of-the-art designs. Thanks to the proposed RA, this work realizes fast and high-linearity residue amplification with superior energy efficiency. The combined AZ and system-level chopping technique effectively mitigate flicker noise without additional thermal noise penalty. The prototyped ADC realizes a Schreier FoM of 183.0dB, corresponding to the best energy efficiency among all published ADCs with >5MS/s conversion rate.

Acknowledgements This work is supported in part by National Key R&D Program of China (2022YFB4401900) and the NSFC (62274005). The corresponding author is Xiyuan Tang.

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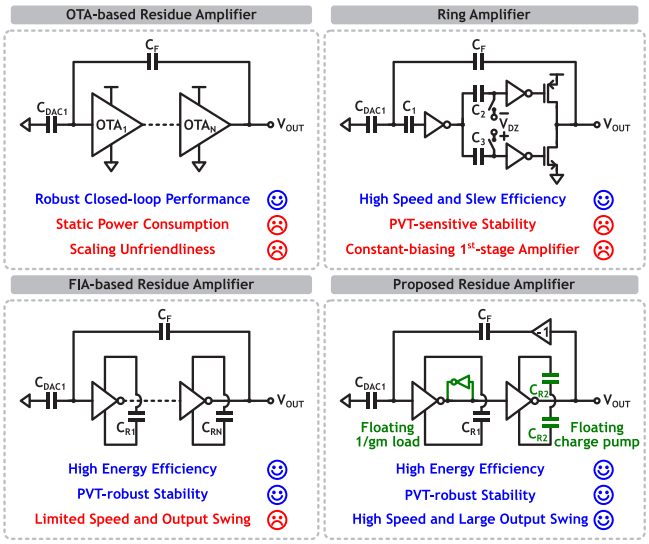


Fig. 1. OTA-based residue amplifier, ring amplifier, FIA-based residue amplifier, and the proposed residue amplifier.

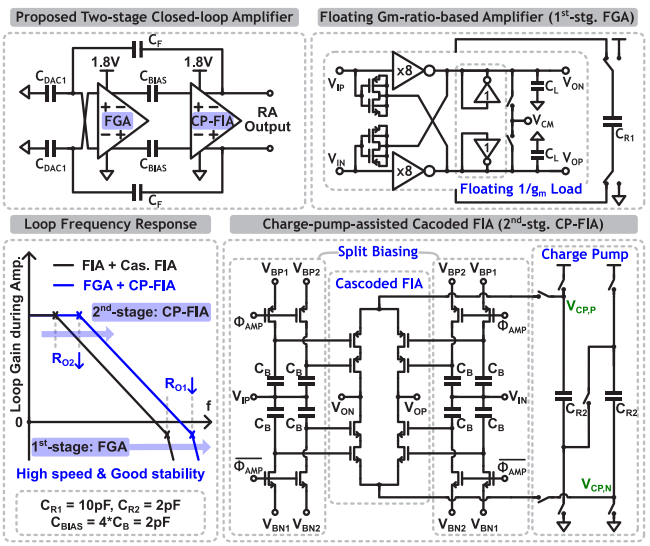


Fig. 2. Schematic of the proposed two-stage closed-loop amplifier with 1st-stage FGA and 2nd-stage CP-FIA.

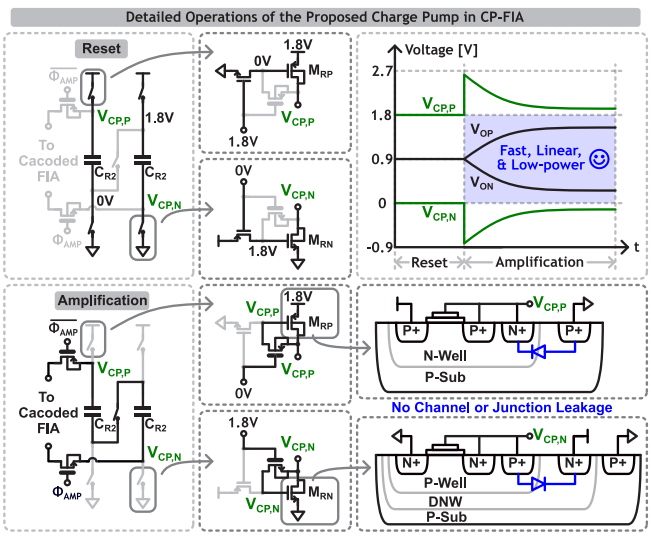


Fig. 3. Detailed operations of the proposed charge pump in CP-FIA and the beyond-the-rail voltage-tolerant switches.

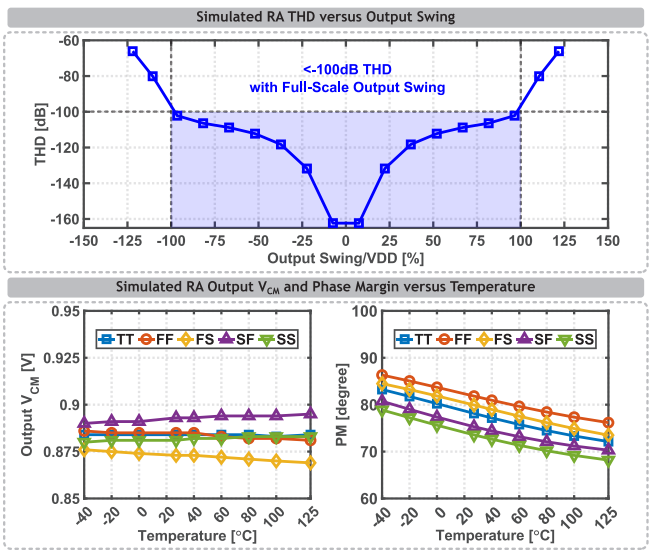


Fig. 4. Simulated RA THD versus output swing, output V_{CM} versus temperature, and phase margin versus temperature.

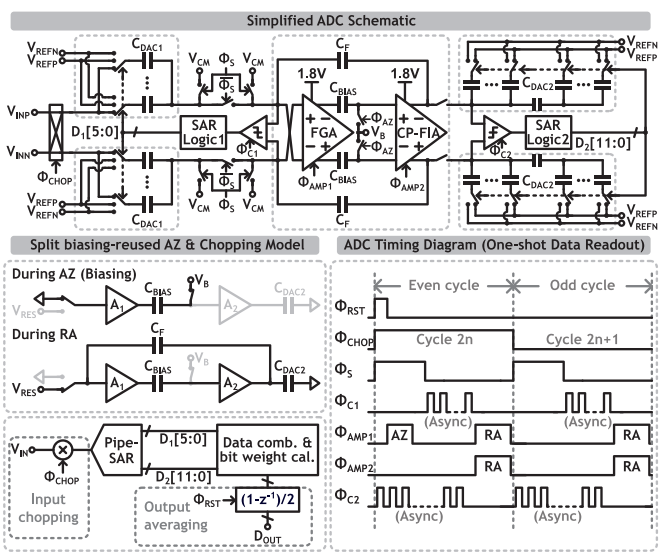


Fig. 5. Simplified ADC schematic, split biasing-reused AZ, ADC chopping model, and timing diagram.

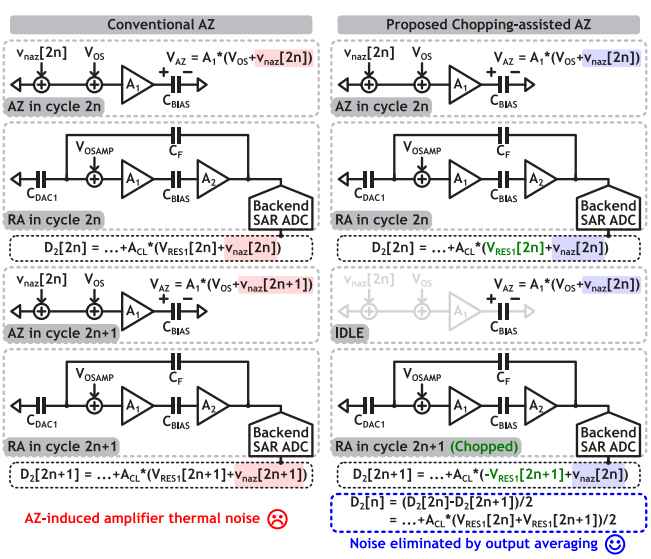


Fig. 6. Operations of conventional AZ and proposed system-level chopping-assisted AZ.

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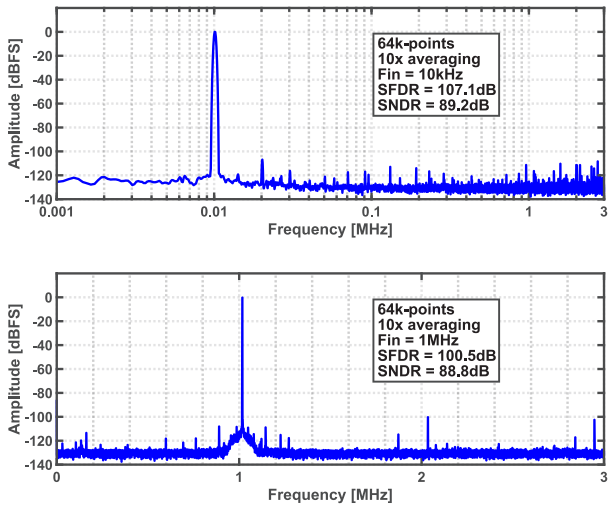


Fig. 7. Measured output spectra with 10kHz input using audio signal source and 1MHz input using waveform generator.

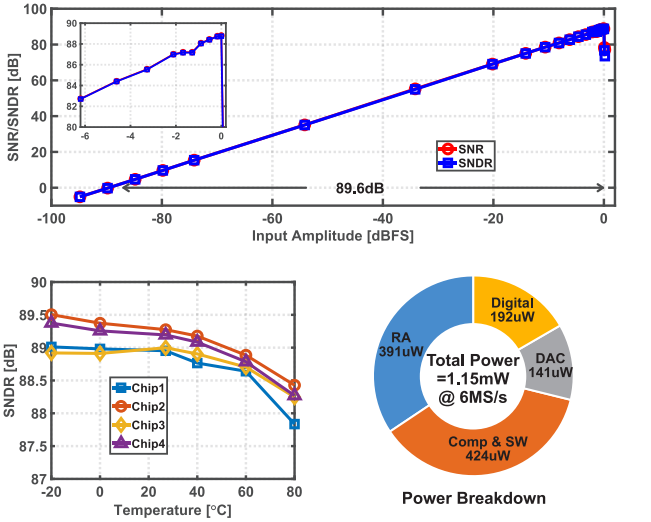


Fig. 8. Measured dynamic range, SNDR variations of 4 chips with temperature variations and power breakdown.

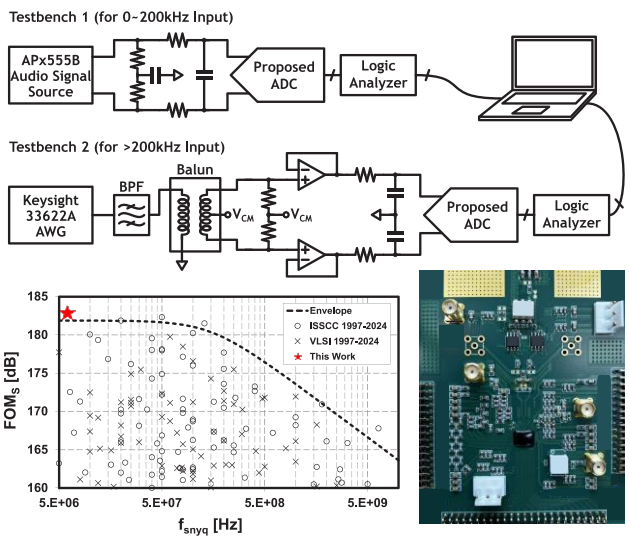


Fig. 9. Testbenches, performance survey, and PCB photo.

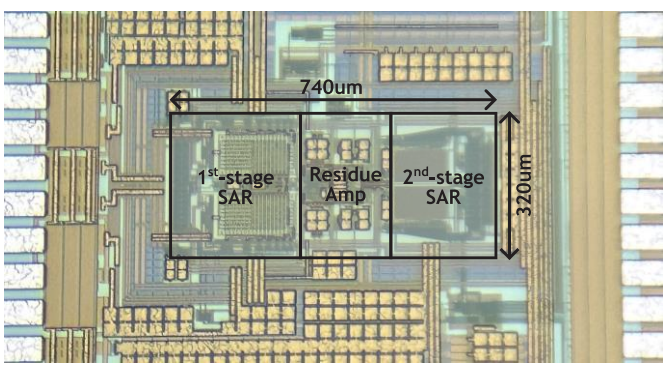


Fig. 10. Chip micrograph.

TABLE I. Comparison with state-of-the-art ADCs

	VLSI-14 Bannon	VLSI-17 Hummerston	ISSCC-24 Bodnar	ISSCC-23 Li	ISSCC-21 Tang	This work
Architecture	Pipelined SAR	Pipelined SAR	Pipelined SAR	Pipelined SAR	Pipelined SAR	**Pipelined SAR**
Residue Amplifier	Closed-loop OTA	Closed-loop OTA	Closed-loop OTA	Ring Amplifier	3-stage FIA	**FGA+ CP-FIA**
Process [nm]	180	180	40	180	40	**180**
Speed [MS/s]	5	2	40	12	40	**6**
Supply [V]	5/1.8	5/1.8	3.3/1.1	3.3/1.8	1.2	**1.8/1.2**
Area [mm^2]	5.74	3.87	0.9	0.6	0.056	**0.237**
Power [mW]	30.52	11.35	50	9.43	0.82	**1.15**
SNDR(SNR*) [dB]	98.6 @1kHz	100* @100kHz	93.7* @1kHz	91.3 @4MHz	75.7 @18MHz	**88.8 @1MHz**
DR [dB]	100.2	100.5	94.2	94.1	79.5	**89.6**
FoMw[1] [fJ/c-s]	87.7	69.4	31.6	26.2	4.1	**8.5**
FoMs[2] [dB]	177.7	179.5	179.7	179.3	179.6	**183.0**

[1]FoMw = Power/(2^{ENOB}·2·BW)
[2]FoMs = SNDR+10·log$_{10}$(BW/Power)

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